Continuity

ANSWER KEY

Section 1: Definition of Continuity at a Point

1 Which three conditions must be satisfied for a function f to be continuous at $x = c$?

a. $f(c)$ **is defined,** $\lim_{x \to c} f(x)$ **exists, and** $\lim_{x \to c} f(x) = f(c)$ ✓

b. $f'(c)$ exists, $f(c) > 0$, and $\lim_{x \to c} f(x) = 0$

c. $\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x)$ only

d. $f(c) = 0$ and $f'(c) = 0$

The formal 3-part definition of continuity at a point requires $f(c)$ is defined, the two-sided limit exists, and the limit equals the function value.

2 Determine if $f(x)$, defined as $\frac{x^2 - 4}{x - 2}$ for $x \neq 2$ and $f(2) = 5$, is continuous at $x = 2$.

a. **Discontinuous (removable discontinuity at** $x = 2$**)** ✓

b. Continuous at $x = 2$

c. Discontinuous (jump discontinuity)

d. Discontinuous (infinite discontinuity)

$\lim_{x \to 2} \frac{(x-2)(x+2)}{x-2} = 4$. However, $f(2) = 5 \neq 4$. The limit exists but differs from $f(2)$, so it is a removable discontinuity.

3 For what value of a is $h(x)$, defined as $ax^2 + 1$ for $x \leq 2$ and $3x - a$ for $x > 2$, continuous everywhere?

a. 1 ✓

b. 2

c. $3/5$

d. -1

Continuity at $x = 2$: $a(2)^2 + 1 = 3(2) - a \Rightarrow 4a + 1 = 6 - a \Rightarrow 5a = 5 \Rightarrow a = 1$.

Section 2: Classification of Discontinuities

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4 Classify the discontinuity of $g(x) = \frac{x-1}{|x-1|}$ at $x = 1$.

a. **Jump discontinuity ✓**

b. Removable discontinuity

c. Infinite discontinuity

d. Essential oscillating discontinuity

$\lim_{x \to 1^+} g(x) = 1$ while $\lim_{x \to 1^-} g(x) = -1$. Because both one-sided limits exist but are unequal, it is a jump discontinuity.

5 Classify the discontinuity of $f(x) = \frac{1}{(x-4)^2}$ at $x = 4$.

a. **Infinite discontinuity ✓**

b. Removable discontinuity

c. Jump discontinuity

d. Continuous

As $x \to 4$, $f(x) \to +\infty$. The function grows without bound, which defines an infinite discontinuity (vertical asymptote).

6 How can $f(x) = \frac{\sin(3x)}{x}$ be defined at $x = 0$ to make it continuous?

a. **Set** $f(0) = 3$ ✓

b. Set $f(0) = 1$

c. Set $f(0) = 0$

d. It cannot be made continuous

Since $\lim_{x \to 0} \frac{\sin(3x)}{x} = 3$, setting $f(0) = 3$ removes the discontinuity, making f continuous.

Section 3: Intermediate Value Theorem (IVT)

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7 State the primary hypothesis required to apply the Intermediate Value Theorem to f on $[a, b]$.

a. f **must be continuous on the closed interval** $[a, b]$ ✓

b. f must be differentiable on (a, b)

c. $f(a) = f(b)$

d. $f'(x) > 0$ on $[a, b]$

The IVT states that if f is continuous on $[a, b]$ and d is between $f(a)$ and $f(b)$, there exists at least one $c \in (a, b)$ such that $f(c) = d$.

8 Use the IVT to show that $x^3 + 2x - 5 = 0$ has a root in which interval?

a. $(1, 2)$ ✓

b. $(0, 1)$

c. $(2, 3)$

d. $(-1, 0)$

Let $f(x) = x^3 + 2x - 5$. $f(1) = 1 + 2 - 5 = -2 < 0$. $f(2) = 8 + 4 - 5 = 7 > 0$. Since f is continuous and changes sign, a root exists in $(1, 2)$.

9 A continuous function g satisfies $g(0) = -3$ and $g(4) = 5$. Does $g(x) = \pi$ have a solution on $(0, 4)$?

a. **Yes, because** $-3 < \pi < 5$ **and** g **is continuous** ✓

b. No, because π is irrational

c. Only if g is differentiable

d. Cannot be determined without $g'(x)$

Since $-3 < \pi \approx 3.1416 < 5$ and g is continuous on $[0, 4]$, the IVT guarantees at least one $c \in (0, 4)$ with $g(c) = \pi$.

Section 4: Extreme Value Theorem (EVT) and Properties

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10 What does the Extreme Value Theorem guarantee for a function continuous on a closed bounded interval $[a, b]$?

a. f **attains both an absolute maximum and an absolute minimum on** $[a, b]$ ✓

b. f has at least one inflection point

c. f is invertible on $[a, b]$

d. $f'(c) = 0$ for some $c \in (a, b)$

The EVT guarantees that any continuous function on a closed bounded interval $[a, b]$ attains both an absolute maximum and an absolute minimum.

11 Which function fails to achieve an absolute maximum on $(0, 1)$?

a. $f(x) = \frac{1}{x}$ ✓

b. $f(x) = x^2$

c. Both $f(x) = 1/x$ and $f(x) = x^2$

d. $f(x) = 5$

On open interval $(0, 1)$, $1/x \to \infty$ (unbounded) and $x^2 < 1$ never reaches 1. Both fail the EVT because the interval is open, not closed.

12 If f and g are continuous at $x = c$, which of the following is NOT guaranteed to be continuous at $x = c$?

a. $f(x)/g(x)$ ✓

b. $f(x) + g(x)$

c. $f(x) \cdot g(x)$

d. $[f(x)]^2$

The quotient f/g is only continuous at c if $g(c) \neq 0$. If $g(c) = 0$, f/g is undefined and discontinuous.

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